V1 master diagram

AI proposes | Script, no AI | You decide | Saved data

Four engines, each with its own steps from start to finish. Some cards repeat so each engine reads alone. The AI never saves anything by itself. It suggests, a script checks, and you confirm.

HOW THE ENGINES CONNECT

1 CAPTURE makes a note → **2 RELATE** links that note to others → **3 MAINTAIN** flags notes that need a recheck → **4 QUERY** reads it all out

All four share one place where notes are saved: a SQLite database, a single file on your computer. Engines 1 to 3 save into it. Engine 4 only reads from it. The database also blocks invalid saves, as a second check after the scripts.

1 CAPTURE

Gets new knowledge in. Ends with one saved note. The 4 kinds of note are at the bottom.

1 You bring a source
A chat or a document you want notes from.

2 Script scans it
Looks for passwords and keys, and warns you.

3 Script saves it
Keeps a full copy, so you can check the source later.

4 AI: worth keeping?
Is it a lasting fact or just chat? Chat is skipped and listed in a log.

5 AI picks the type
Decision = a choice. Assumption = a guess. Finding = a discovery. Question = unknown.

6 AI writes the note
One clear sentence saying what it means, and where it came from.

7 Script checks it
Is the type allowed? Is the quote really in the source? Is it almost a copy of an old note?

8 You confirm
Nothing is saved until you say yes.

THE 4 KINDS OF NOTE

Decision
A choice the project made, with the reason why. Example: use dark green buttons.

Assumption
Something we treat as true but have not proven. Example: users have fast phones.

Finding
Something we discovered, with its source. Example: blue failed the contrast test.

Question
Something we do not know yet. Example: does dark green pass the test?

Status of every note
Active = still counts. Challenged = needs a second look. Closed = finished, with a reason.

TECH BRIEF

- **AI model:** a language model reads the source and proposes notes. It can only act through scripts, so it never writes to the database itself.
- **Secret scan:** a script looks for obvious keys, passwords and tokens in the text before the source is saved.
- **Quote check:** if a note carries a quote, a script checks that exact text is in the saved source. It proves the quote is there, not that the note follows from it.
- **Duplicate warning:** a plain text comparison with Active and Challenged notes. It warns, and you choose to save or cancel. No AI.
- **Weak spot:** capture quality is the main risk. Only a clean design chat was tested so far. A messy real chat is still untested.

2 RELATE

Connects the new note to notes it belongs with. Ends with one saved link. The 5 kinds of link are at the bottom.

1 A new note is saved
This comes from Capture.

2 Script finds matches
Picks old notes that sound similar or are already connected. No AI.

3 AI picks pairs
It only suggests which notes go together. It does not choose how they connect.

4 Script sets the link kind
From the two note types. A Decision next to a Question can only be 'answers'.

5 You choose, if needed
Only when more than one kind fits. The script asks you. The AI never picks the kind.

6 You accept or reject
Say yes or no to each link. A no is remembered, so it is not suggested again.

7 Script saves the link
Then Maintain checks if anything else must change.

THE 5 KINDS OF LINK

supports
A Finding backs up a Decision or an Assumption.

contradicts
A Finding goes against a Decision, an Assumption or another Finding.

depends_on
A Decision rests on an Assumption, a Finding or another Decision. An Assumption rests on a Finding or an Assumption.

supersedes
A new note replaces an old note of the same type. The old one is closed.

answers
A Finding or a Decision answers a Question. The Question is closed.

TECH BRIEF

- **Embedding model (bge-base-en-v1.5):** turns each note into a list of numbers that captures its meaning. Notes with close numbers are similar in meaning. The 10 closest become the matches.
- **Not keyword search:** BM25 (keyword ranking) is no longer used to find matches. In the test it found 0 of 16 links between notes that share no words. The embedding model found 25 of 32 checks in its top 10.
- **Other signals:** notes already linked, same project area, and time. A script reads them from the saved fields.
- **Link kind:** a script works out the kind from the two note types. If more than one kind is allowed, it picks none and asks you. The AI never chooses the kind.
- **Link rules:** a script refuses links to itself, duplicate links and loops. The database repeats these checks.
- **Weak spot:** the test used only 38 notes. It must be run again at 500 to 5,000 notes.

3 MAINTAIN

Keeps notes true when something changes. No AI. Only directly connected notes are marked, never further.

1 Something changes
You accepted a link, or you closed a note.

2 Script finds the rule
Checks the 4 rules below. No AI is used.

3 Script changes notes
Closes a note, or marks one 'Challenged', which means it needs a second look.

4 You review
You decide what to do with each Challenged note.

THE 4 RULES

Rule 1: replaced
You accept a link saying a new note replaces an old one. The old note is closed.

Rule 2: knock-on
A note is closed as wrong, replaced or withdrawn. Notes that relied on it become Challenged.

Rule 3: conflict
You accept a link saying two notes disagree. The note it points at becomes Challenged. You choose which one it points at.

Rule 4: answered
You accept a link saying a note answers a Question. The Question is closed.

TECH BRIEF

- **Fixed rules in code:** 4 triggers that run inside add_link and close_item. No AI and no guessing.
- **All or nothing:** each save is one database transaction. If closing a note fails halfway, nothing is changed.
- **Change log:** every change adds one row: what, old value, new value, date and cause. Rows are never edited or deleted.
- **One step only:** a Challenged note does not flag the notes that depend on it. Cost: second-level notes may be trusted wrongly until you recheck.

4 QUERY

Reads notes out. Read only. Changes nothing.

1 You or an AI chat asks
At any time.

2 You pick a way to ask
The 4 ways are listed below.

3 Script reads the notes
It only looks. It never changes anything.

4 A list comes back
The saved notes and links as they are. The AI does not rewrite them.

THE 4 WAYS TO ASK

Current state
All notes that still count. Challenged ones come first.

Topic search
Type a word. Notes that use it come first.

Why
Follow the links from a note to see the reasons behind it.

History
See what changed, the old and new text, the date and why.

TECH BRIEF

- **Read only:** the query action cannot change anything. It returns the stored notes and links as they are.
- **Topic search:** this is where BM25 is still used. It is a standard way to rank text by matching words. No AI in the ranking. SQLite has a built-in search that could do this, not yet checked.
- **Live reading:** an AI chat can call Query itself during a session. It can read, never write.
- **Text export:** a script writes the notes into a readable text file, so you can read them without a database tool.

1 Capture

Same chain as on the master diagram.

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